

Quantifying the Affective Gap: A Zero-Shot Evaluation of LLMs on Fine-Grained Emotion Taxonomies

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Abstract—Emotion recognition in natural language is a foundational challenge in affective computing, with critical implications for human-computer interaction, mental health support, and conversational AI. This paper presents a rigorous, unified zero-shot evaluation of three leading commercial large language models: Claude (claude-sonnet-4-6), ChatGPT (GPT-5.4), and Gemini (gemini-2.5-flash). The models were queried through their respective production APIs as of April 2026 on a fine-grained 13-class emotion classification task. Using a stratified 1,000-sentence sample from the *boluix/emotions-dataset* comprising 131,306 sentences across 13 categories, a single uniform prompt with no exemplars was applied identically across all models. Gemini achieves the highest accuracy (39.9%) and macro-F1 (0.363), followed by GPT-5.4 (38.8%, F1=0.291) and Claude (38.0%, F1=0.159). All models excel on *sarcasm* and *desire* while consistently failing on *love*, *confusion*, and *shame*. McNemar’s tests reveal no statistically significant pairwise differences ($p > 0.10$), suggesting convergence at a shared zero-shot ceiling. Claude’s markedly lower macro-F1 exposes a class-imbalance prediction bias. These findings highlight the current limitations of frontier AI systems in zero-shot fine-grained emotion classification.

Index Terms—emotion classification, affective computing, LLMs, zero-shot learning, NLP, GPT-5.4, Claude, Gemini, AI, human-computer interaction

I. INTRODUCTION

Emotions shape human communication and underpin high-stakes applications including mental health monitoring [1], empathetic dialogue systems [2], and human-computer interaction. The ability of an AI system to recognize affective states is therefore a prerequisite for deployment in any domain where human welfare is at stake.

Despite rapid advances in large language models (LLMs), their affective intelligence remains poorly characterized at fine-grained resolution. Most studies probe coarse polarity or the Ekman six basic emotions [3] rather than richer taxonomies [4], [5], and direct cross-provider comparisons under identical experimental conditions are absent from the literature. This gap matters: practitioners selecting an API for emotion-aware applications must rely on ad hoc or proprietary benchmarks that may not reflect real-world linguistic diversity.

This paper addresses the gap through four research questions: zero-shot accuracy across 13 classes (RQ1); pairwise

statistical differences using McNemar’s test (RQ2); per-class strengths and failure modes (RQ3); and the effect of sentence length on accuracy (RQ4). We contribute (i) the first direct zero-shot comparison of Claude (claude-sonnet-4-6), ChatGPT (GPT-5.4), and Gemini (gemini-2.5-flash) on a 13-class task using production APIs as of April 2026; (ii) per-emotion accuracy breakdowns revealing systematic failure modes; (iii) McNemar significance testing; and (iv) a sentence-length moderation analysis.

II. RELATED WORK

A. Emotion Classification and Datasets

Computational emotion recognition has progressed from lexicon-based resources such as the NRC Lexicon [6] and SemEval affective tasks [7]–[9] through deep learning classifiers to transformer fine-tuning paradigms achieving state-of-the-art results on benchmarks including GoEmotions [10] (58K comments, 27 categories). BERT [11], [12] and RoBERTa [13] established the transformer paradigm as dominant for supervised emotion classification. The *boluix/emotions-dataset* extends the Ekman taxonomy with social (shame, guilt) [14], cognitive (confusion), rhetorical (sarcasm) [15], and motivational (desire) categories, offering broader ecological validity.

Emotion-aware AI may also complement multimodal assistive systems that integrate cognitive, physiological, and attentional context in knowledge-work environments [16]. Related human-computer interaction research has examined how eye-tracking signals can characterize variations in cognitive effort during digital-document reading [17]–[19].

B. LLMs for Emotion and Sentiment

GPT-3 [20] demonstrated zero-shot classification through in-context learning, with instruction tuning [21], [22] substantially improving generalization. SentimentGPT [5] found LLMs competitive on coarse polarity but weaker on low-frequency emotion classes. Yang et al. [4] evaluated ChatGPT and GPT-4 on sentiment tasks but excluded Claude, Gemini, and a fine-grained 13-class taxonomy the gap this study directly addresses.

TABLE I
CLASS DISTRIBUTION IN STRATIFIED SAMPLE ($N = 1,000$)

Emotion	Count	%	Emotion	Count	%
happiness	205	20.5	disgust	60	6.0
neutral	135	13.5	surprise	44	4.4
sadness	130	13.0	shame	34	3.4
anger	107	10.7	sarcasm	26	2.6
love	76	7.6	guilt	26	2.6
confusion	70	7.0	desire	19	1.9
fear	68	6.8			
Total: 1,000 (100.0%)					

III. METHODOLOGY

A. Dataset

The *boluix/emotions-dataset* [23] is a publicly available Hugging Face corpus of 131,306 English sentences annotated with one of 13 mutually exclusive emotion labels: *happiness*, *sadness*, *fear*, *anger*, *love*, *disgust*, *surprise*, *neutral*, *confusion*, *desire*, *shame*, *guilt*, and *sarcasm*. A stratified random sample of $N = 1,000$ sentences was drawn with seed = 42, preserving class proportions to simulate realistic deployment conditions. Table I reports the distribution. *Happiness* is the most frequent class (20.5%, $n = 205$); *desire* is the rarest (1.9%, $n = 19$). This 10:1 ratio has direct consequences for macro-F1: models concentrating predictions on high-frequency classes achieve deceptively high accuracy while failing on minority classes.

B. Models and API Configuration

Three frontier LLMs were queried through their production APIs as of April 2, 2026: Claude (claude-sonnet-4-6) through Anthropic’s Messages API with default sampling; ChatGPT (GPT-5.4) through OpenAI’s Chat Completions API with temperature=0; and Gemini (gemini-2.5-flash) through Google’s Generative AI API with default sampling. All models received max_tokens=10, enforcing single-word responses. Responses not matching one of the 13 canonical labels after case normalization were treated as errors; all three models exhibited strong label adherence in practice.

C. Prompt Design

A single-turn prompt was used identically across all models, with no system message and no few-shot examples [20], [24]: “Classify the emotion expressed in this sentence. Reply with ONLY one word from this list: happiness, sadness, fear, anger, love, disgust, surprise, neutral, confusion, desire, shame, guilt, sarcasm. Sentence: {sentence}. Emotion:”

The full label list constrains the output space; imperative instruction style elicits consistent outputs from instruction-tuned models [21]. No chain-of-thought preamble is included, ensuring performance reflects raw affective inference [24].

D. Evaluation Metrics

Four metrics were computed per model: (i) **Overall Accuracy**; (ii) **Macro-F1** ($F1_{\text{mac}}$), the primary metric for minority-class evaluation; (iii) **Weighted-F1** ($F1_{\text{wt}}$); and (iv) **Cohen’s**

TABLE II
OVERALL PERFORMANCE ($N = 1,000$)

Model	Acc. (%)	$F1_{\text{mac}}$	$F1_{\text{wt}}$	κ
Claude (sonnet-4-6)	38.0	0.159	0.394	0.320
GPT-5.4	38.8	0.291	0.387	0.327
Gemini (2.5-flash)	39.9	0.363	0.394	0.337

TABLE III
PER-EMOTION ACCURACY (%) BY MODEL

Emotion	Claude	GPT-5.4	Gemini
sarcasm	96.2	100.0	100.0
desire	89.5	94.7	94.7
anger	53.3	61.7	61.7
guilt	50.0	42.3	34.6
fear	44.1	47.1	48.5
sadness	43.1	55.4	49.2
happiness	35.6	24.9	45.9
disgust	30.0	35.0	30.0
neutral	31.1	25.9	14.8
shame	29.4	17.6	26.5
surprise	27.3	25.0	27.3
confusion	22.9	25.7	27.1
love	14.5	27.6	14.5
Overall	38.0	38.8	39.9

Kappa (κ) [25]. Pairwise comparisons used **McNemar’s test** [26] with continuity correction.

IV. RESULTS

A. Overall Performance

Table II summarizes aggregate performance. Gemini leads on all four metrics (accuracy 39.9%, macro-F1 0.363, $\kappa = 0.337$), followed by GPT-5.4 (38.8%, 0.291, 0.327) and Claude (38.0%, 0.159, 0.320). All three models substantially exceed the majority-class baseline (20.5%) and random baseline (7.7%), yet fall far below human-level performance (>70%) [10]. The narrow 1.9 pp spread suggests convergence near a shared zero-shot ceiling, confirmed by the McNemar results below.

B. Per-Emotion Accuracy

Table III and Fig. 2 present per-class accuracy. *Sarcasm* (96.2%–100%) and *desire* (89.5%–94.7%) stand in stark contrast to all other categories. *Anger* (53.3%–61.7%), *fear* (44.1%–48.5%), and *sadness* (43.1%–55.4%) form a mid-performance cluster. *Love* (14.5%–27.6%), *confusion* (22.9%–27.1%), and *shame* (17.6%–29.4%) are consistently the hardest categories. Notable model-specific divergences include Gemini collapsing on *neutral* (14.8% vs. Claude’s 31.1%), Claude leading on *guilt* (50.0%) and *neutral*, and GPT-5.4 leading on *sadness* and *love*.

Fig. 1 presents normalized confusion matrices. Cross-model patterns include *love* misclassified as *happiness*, confirming semantic overlap between positive affect categories; *confusion* misclassified as *neutral* and *sadness*; and *fear* frequently

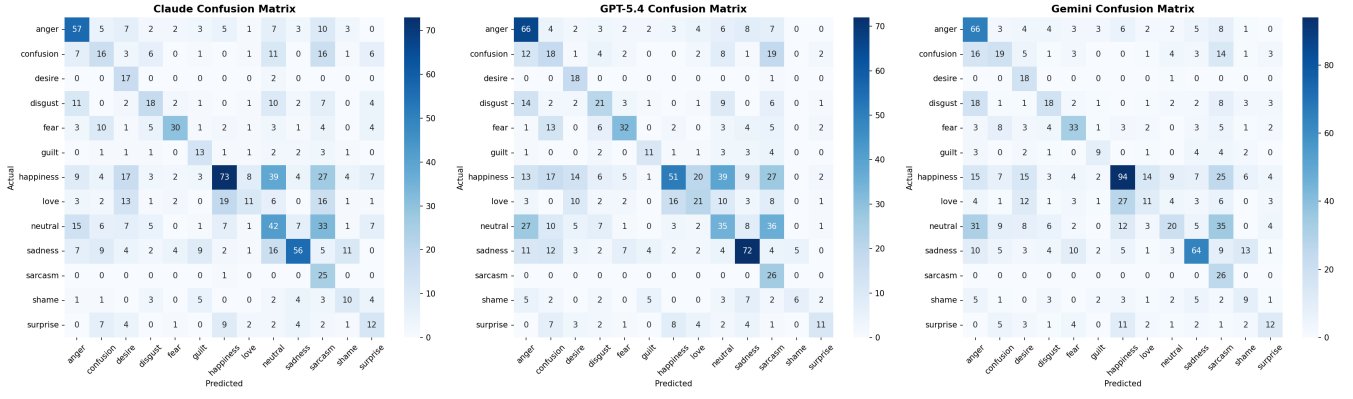


Fig. 1. Normalized confusion matrices for Claude, GPT-5.4, and Gemini. Off-diagonal mass concentrates on semantically proximal pairs: *love* ↔ *happiness*, *fear* ↔ *sadness*, and *confusion* ↔ *neutral*.

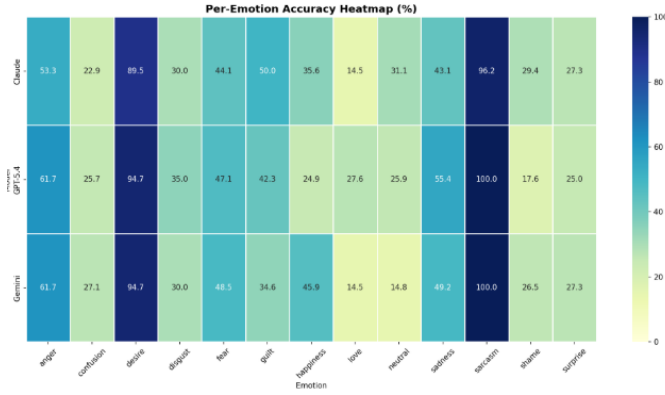


Fig. 2. Per-emotion accuracy heatmap (%). Sarcasm scores highest (96.2–100%); love and neutral are consistently weakest.

TABLE IV
MCNEMAR’S TEST p -VALUES ($\alpha = 0.05$)

Comparison	p -value	Significant?
Claude vs. GPT-5.4	0.5083	No
Claude vs. Gemini	0.1350	No
GPT-5.4 vs. Gemini	0.4095	No

assigned to *sadness*, suggesting that shared negative valence dominates arousal-level distinctions.

C. Statistical Significance

Table IV presents McNemar’s test p -values. No comparison reaches significance at $\alpha = 0.05$, supporting convergence: accuracy differences are consistent with random variation, and no model can be declared definitively superior under this protocol.

D. Sentence Length Analysis

Performance peaks in the medium range (5–15 words): Claude=41.3%, GPT-5.4=42.0%, and Gemini=43.6% ($n = 567$). Long sentences (>15 words) produce the lowest accuracy (33.2%–34.3%), a ~ 9 pp drop consistent across all

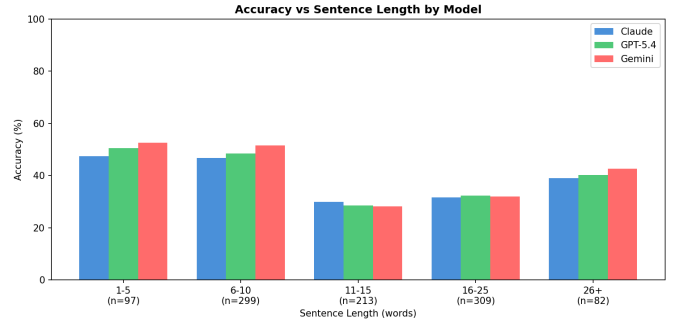


Fig. 3. Accuracy by sentence length. Medium sentences (5–15 words) yield peak performance; long sentences (>15 words) degrade accuracy by ~ 9 pp across all models.

models (Fig. 3), suggesting a structural property of zero-shot affective inference.

V. DISCUSSION

Zero-shot ceiling: The 1.9pp accuracy spread and non-significant McNemar results indicate all three models have converged near a shared zero-shot ceiling. For practitioners, provider choice is unlikely to affect raw accuracy. However, the macro-F1 gap (0.159 vs. 0.363, Claude vs. Gemini) is practically significant: in applications requiring minority-class sensitivity (e.g., shame, guilt), Gemini is the preferred choice.

Sarcasm/desire paradox and hard categories: Near-perfect sarcasm (96.2%–100%) and desire (89.5%–94.7%) accuracies likely reflect small support sizes ($n = 26$, $n = 19$) and atypical lexical prototypicality [27] rather than robust affective understanding. Conversely, *love* is the worst-performing class for Claude and Gemini (14.5%), driven by semantic overlap with *happiness*. The confusion matrix confirms that most *love* misclassifications land on *happiness*, consistent with GoEmotions [10], where positive social emotions formed the hardest discrimination cluster.

Provider-specific biases: Claude’s macro-F1 of 0.159 less than half of Gemini’s 0.363 despite comparable accuracy (38.0%) is the signature of majority-class prediction bias [22].

Claude over-predicts *happiness* while underperforming on minority classes, likely amplified by alignment processes during pretraining [21]. This is a critical limitation for mental health applications where rare emotions carry high clinical significance. Gemini, by contrast, collapses on *neutral* (14.8% vs. Claude’s 31.1%), systematically reassigning neutral sentences to emotional categories. This pattern may reflect RLHF objectives rewarding emotionally engaging outputs [21], making Gemini unreliable for applications requiring neutral-state detection.

Implications and limitations: Accuracy in the 38%–40% range falls well short of deployment thresholds for emotionally sensitive contexts [1], [2]. Key limitations include the zero-shot-only evaluation (few-shot prompting may substantially improve all models), reliance on a single English-language dataset, and high-variance estimates for rare classes ($n < 30$). The results reflect API behavior as of April 2, 2026; continuous model updates may alter findings.

VI. CONCLUSION

We presented the first direct zero-shot evaluation of Claude, GPT-5.4, and Gemini on 13-class emotion classification using production APIs. Gemini led on all metrics (accuracy 39.9%, macro-F1 0.363), although no pairwise difference was statistically significant, suggesting convergence near a shared zero-shot ceiling. Our results reveal that frontier LLMs remain substantially below human-level affective awareness in zero-shot fine-grained settings. Future work should explore few-shot and chain-of-thought prompting [24] for minority-class improvement, fine-tuned baselines to quantify the zero-shot gap, and multilingual evaluation to probe cross-lingual affective generalization.

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